

Comparative Analysis of Numerical Schemes **and** **Physics-Informed Neural Networks** for Two-Dimensional Dam-Break Problems with Source Terms under Different Initial Dam Breaking Conditions

Zahra¹, **Michael Ajao-Olarinoye**², M. Asif Farooq¹, and Firdos Khan¹

¹Department of Mathematics, School of Natural Sciences (SNS), National University of Sciences and Technology (NUST), Islamabad, Pakistan

zahasamad1998@gmail.com (zahra), asiffarooq.2007@gmail.com (M. Asif Farooq), fkyousafzai@gmail.com (Firdos Khan)

²**Centre for Computational Science and Mathematical Modelling, Coventry**

University, Coventry, United Kingdom

ajaoolarinoyemichael@gmail.com

July 10, 2026

1 Introduction

The catastrophic failure of dams releases large volumes of water that propagate downstream as rapidly varying, highly unsteady free-surface flows, posing severe risks to infrastructure and hu-

man life. Accurate and fast simulation of dam-break wave propagation is therefore a cornerstone of hydraulic engineering, flood hazard assessment, and emergency response planning. The hydrodynamics of such events are described by the two-dimensional shallow water equations (SWEs), a nonlinear hyperbolic system obtained by depth-averaging the Navier–Stokes equations under a hydrostatic pressure assumption [1, 2]. A defining feature of this system is that discontinuities (shocks, hydraulic jumps) form in finite time even from smooth initial data, and realistic modeling must additionally account for bed-topography source terms and frictional resistance [3, 4].

For decades the standard tools for these problems have been shock-capturing finite difference and finite volume schemes. This family includes the classical Lax–Wendroff method [5], Godunov-type methods built on exact or approximate Riemann solvers [6–9], and higher-order MUSCL-type reconstructions with slope limiters [10] advanced by strong-stability-preserving time integrators [11]. These methods offer rigorous conservation and robust shock capturing, but their cost scales with grid resolution and the CFL constraint, which limits their use in real-time forecasting and ensemble-based risk assessment.

Scientific machine learning offers a complementary paradigm. Physics-informed neural networks (PINNs) [12] parameterize the space-time solution with a neural network trained on the PDE residual, and have been applied to the SWEs in a growing body of work [13–17]. However, strong-form PINNs face well-documented difficulties on hyperbolic problems. The pointwise residual is ill-defined at discontinuities [18, 19], the trained network does not enforce discrete conservation and can leak mass, and spectral bias drives the solution toward an overly smooth, dissipative approximation of sharp fronts [20]. Hybrid approaches that replace the strong-form residual with a discrete finite-volume residual restore conservation structure [21–25], but their behavior across systematically varied dam-break initial conditions and topography source terms remains largely unexplored.

Several recent studies have compared neural solvers against classical schemes for dam-break flows. Mumtaz et al. [16] compared PINNs against a Lax–Wendroff baseline for 1D and 2D dam-break scenarios and found that PINNs reproduce the reference solutions with limited accuracy, improving when numerical solutions are injected into training. Tian et al. [15] benchmarked PINNs

for the 2D SWEs with topography and rainfall source terms against an HLL scheme, finding errors concentrated at shock and rarefaction fronts. Qi et al. [14] compared data-free PINNs against a finite-volume solver, including a real-world flood event. Liu [21] showed that a finite-volume-informed PINN loss produces a rugged landscape in which physics-only training collapses to a spurious low-momentum state unless sparse data guidance is supplied.

2 Problem Setup

In this section, the two dimensional shallow water equations (SWEs) and the numerical schemes used to solve them are presented. The governing equations used in this study are the same as the standard equations, but they include bed topography source terms. The reference solutions are computed using the three numerical schemes: the Lax–Wendroff (LW) scheme, the Harten–Lax–van Leer (HLL) approximate Riemann solver, and the MUSCL-Rusanov (MUSCL-RS) scheme. The six dam-break scenarios examined include Step, Rectangular, Circular, Gaussian, Parabolic and Triangular water height profiles. These benchmark problems are described in detail in Sect. 2.3. The numerical solutions are then evaluated and compared with respect to the performance, accuracy and shock capturing ability of the numerical schemes considered. Section 2.4 then introduces the scientific machine learning models evaluated against these reference solutions. These comprise strong-form physics-informed neural networks in primitive and conservative formulations, together with a finite-volume-informed PINN trained on the discrete residual of a differentiable HLLC solver step.

2.1 Governing Equations

The mathematical model governing the hydrodynamic flow in this study is the two-dimensional (2D) Shallow Water Equations (SWEs). The equations are obtained by hydrostatic pressure distribution, by neglecting the vertical acceleration, and by considering the fluid density as constant and by depth integrating the Navier-Stokes equations. The SWEs are hyperbolic conservation laws which are in effect for the propagation of water waves in domains where the horizontal length scale

is much greater than the water depth. The present study is based on the standard formulation of the equations to which a source term vector $\mathbf{S}$, representing the effects of bottom topography (bed slope), and bed friction are added. These source terms are important to include in order to simulate a realistic flood, as they describe the effects of topography and resistance to flow.

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x, y, t, \mathbf{U}), \quad (1)$$

where t denotes time, x and y are the spatial coordinates, $\mathbf{U}$ is the vector of conserved variables, $\mathbf{F}$ and $\mathbf{G}$ are the flux vectors in the x - and y -directions, respectively, and $\mathbf{S}$ is the source term vector.

The vectors $\mathbf{U}$, $\mathbf{F}$, and $\mathbf{G}$ are defined as

$$\mathbf{U} = \begin{bmatrix} h \\ hu \\ hv \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} hu \\ hu^2 + \frac{1}{2}gh^2 \\ huv \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} hv \\ huv \\ hv^2 + \frac{1}{2}gh^2 \end{bmatrix}. \quad (2)$$

These are the expressions for the water depth h , the depth-averaged velocity components, u and v in the x and y directions, respectively, and g the gravitational acceleration. The quantities hu and hv are the momentum fluxes per unit width.

The source term vector $\mathbf{S}$ represents the forces on the flow other than its internal forces. The influence of bed slope (topography) and bed friction (Manning roughness) are taken into account in the present study. As a result, $\mathbf{S}$ is given by

$$\mathbf{S} = \begin{bmatrix} 0 \\ -gh \frac{\partial Z}{\partial x} - \frac{gn^2 u \sqrt{u^2 + v^2}}{h^{4/3}} \\ -gh \frac{\partial Z}{\partial y} - \frac{gn^2 v \sqrt{u^2 + v^2}}{h^{4/3}} \end{bmatrix}. \quad (3)$$

In this equation $Z(x, y)$ is the bed elevation and n is the Manning roughness coefficient. The

first term of $\mathbf{S}$ is null, meaning that the transfer of mass from an external source (rainfall and/or infiltration) is not taken into account in the continuity equation. The second term and third term are the terms that represent the momentum source terms in the x and y directions, respectively.

The terms $-gh\frac{\partial Z}{\partial x}$, $-gh\frac{\partial Z}{\partial y}$ represent the driving forces due to the sloping bed, where g and h are the magnitude of gravity and the height above sea level, respectively. The terms involving n represent the losses due to the friction between the bed and the water.

For the present numerical study, the Manning coefficient is assumed as $n = 0$. The friction terms then drop off and the source term vector becomes

$$\mathbf{S} = \begin{bmatrix} 0 \\ -gh\frac{\partial Z}{\partial x} \\ -gh\frac{\partial Z}{\partial y} \end{bmatrix}. \quad (4)$$

2.2 Numerical Schemes

In this study, three distinct numerical schemes are employed to solve the two-dimensional shallow water equations with source terms. These schemes are the Lax–Wendroff (LW) method, the Harten–Lax–van Leer (HLL) approximate Riemann solver, and the MUSCL scheme with third-order Strong Stability Preserving Runge–Kutta (SSP–RK3) time integration. Each scheme is described in detail in the following subsections.

2.2.1 Lax–Wendroff Scheme

The Lax–Wendroff scheme is a second-order accurate finite difference method that employs a predictor–corrector approach. It is widely used for solving hyperbolic conservation laws due to its ability to capture wave propagation with reasonable accuracy while maintaining computational efficiency.

The scheme follows a two-step procedure at each time level.

Predictor Step The intermediate solution at the half time level $t^{n+\frac{1}{2}}$ is computed as

$$U_{i,j}^{n+\frac{1}{2}} = U_{i,j}^n - \frac{\Delta t}{2\Delta x} (F_{i+1,j}^n - F_{i-1,j}^n) - \frac{\Delta t}{2\Delta y} (G_{i,j+1}^n - G_{i,j-1}^n) + \frac{\Delta t}{2} S_{i,j}^n, \quad (5)$$

where $U_{i,j}^{n+\frac{1}{2}}$ denotes the predicted conservative variables at the half time level, and $S_{i,j}^n$ represents the source term vector associated with bed topography.

The source term vector is given by

$$S = \begin{bmatrix} 0 \\ -gh \frac{\partial Z}{\partial x} \\ -gh \frac{\partial Z}{\partial y} \end{bmatrix}, \quad (6)$$

where $Z(x, y)$ denotes the bed elevation.

Flux Evaluation After computing the intermediate variables, the flux vectors are evaluated at the half time level as

$$F^{n+\frac{1}{2}} = \begin{bmatrix} hu \\ hu^2 + \frac{1}{2}gh^2 \\ huv \end{bmatrix}, \quad G^{n+\frac{1}{2}} = \begin{bmatrix} hv \\ huv \\ hv^2 + \frac{1}{2}gh^2 \end{bmatrix}. \quad (7)$$

Corrector Step The full time-step update is given by

$$U_{i,j}^{n+1} = U_{i,j}^n - \frac{\Delta t}{\Delta x} (F_{i+\frac{1}{2},j}^{n+\frac{1}{2}} - F_{i-\frac{1}{2},j}^{n+\frac{1}{2}}) - \frac{\Delta t}{\Delta y} (G_{i,j+\frac{1}{2}}^{n+\frac{1}{2}} - G_{i,j-\frac{1}{2}}^{n+\frac{1}{2}}) + \Delta t S_{i,j}^{n+\frac{1}{2}}. \quad (8)$$

The scheme achieves second-order accuracy in both space and time. However, it may produce spurious oscillations near steep gradients or discontinuities, which is a known limitation of this method.

2.2.2 Harten–Lax–van Leer (HLL) Scheme

The HLL scheme is an approximate Riemann solver that computes intercell fluxes based on wave-speed estimates. It is a Godunov-type finite volume method that provides robust solutions for hyperbolic conservation laws, particularly in the presence of shocks and discontinuities.

For the x -direction, the HLL flux between the left and right states, denoted by $\mathbf{U}_L$ and $\mathbf{U}_R$, respectively, is given by

$$\mathbf{F}_{i+\frac{1}{2}}^{HLL} = \begin{cases} \mathbf{F}_L, & \text{if } S_L \geq 0, \\ \frac{S_R \mathbf{F}_L - S_L \mathbf{F}_R + S_L S_R (\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L}, & \text{if } S_L < 0 < S_R, \\ \mathbf{F}_R, & \text{if } S_R \leq 0. \end{cases} \quad (9)$$

where the wave speeds are estimated as

$$S_L = \min(u_L - c_L, u_R - c_R), \quad (10)$$

$$S_R = \max(u_L + c_L, u_R + c_R), \quad (11)$$

where

$$c_L = \sqrt{gh_L}, \quad c_R = \sqrt{gh_R}, \quad (12)$$

and

$$u_L = \frac{(hu)_L}{h_L}, \quad u_R = \frac{(hu)_R}{h_R}. \quad (13)$$

Similarly, for the y -direction, the HLL flux is given by

$$\mathbf{G}_{j+\frac{1}{2}}^{HLL} = \begin{cases} \mathbf{G}_B, & \text{if } S_L \geq 0, \\ \frac{S_R \mathbf{G}_B - S_L \mathbf{G}_T + S_L S_R (\mathbf{U}_T - \mathbf{U}_B)}{S_R - S_L}, & \text{if } S_L < 0 < S_R, \\ \mathbf{G}_T, & \text{if } S_R \leq 0. \end{cases} \quad (14)$$

where the wave speeds in the y -direction are estimated as

$$S_L = \min(v_B - c_B, v_T - c_T), \quad (15)$$

$$S_R = \max(v_B + c_B, v_T + c_T), \quad (16)$$

with

$$c_B = \sqrt{gh_B}, \quad c_T = \sqrt{gh_T}. \quad (17)$$

The source term is incorporated explicitly through the semi-discrete formulation

$$\frac{\partial \mathbf{U}}{\partial t} = -\frac{\partial \mathbf{F}}{\partial x} - \frac{\partial \mathbf{G}}{\partial y} + \mathbf{S}. \quad (18)$$

The spatial derivatives are approximated using the HLL fluxes, whereas the temporal integration is performed using the SSP–RK3 scheme described in the next subsection.

2.2.3 MUSCL Scheme with SSP–RK3 Time Integration

The **Monotonic Upstream-centered Scheme for Conservation Laws (MUSCL)** is a second-order spatial reconstruction technique that improves the accuracy of the finite volume method while preserving monotonicity near discontinuities. In the present work, the MUSCL scheme is coupled with the Minmod slope limiter, the Rusanov (Local Lax–Friedrichs) numerical flux, and the third-order Strong Stability Preserving Runge–Kutta (SSP–RK3) method for time integration.

Spatial Reconstruction Within the finite volume framework, only the cell-averaged conservative variables are stored. Therefore, the left and right states at each cell interface are reconstructed using the MUSCL approach as

$$U_{i+\frac{1}{2}}^L = U_i + \frac{1}{2} \Delta x \Delta U_i, \quad (19)$$

$$U_{i+\frac{1}{2}}^R = U_{i+1} - \frac{1}{2} \Delta x \Delta U_{i+1}, \quad (20)$$

where the limited slope is computed using the Minmod limiter

$$\Delta U_i = \text{minmod} \left(\frac{U_i - U_{i-1}}{\Delta x}, \frac{U_{i+1} - U_i}{\Delta x} \right). \quad (21)$$

The Minmod function is defined as

$$\text{minmod}(a, b) = \frac{1}{2} [\text{sign}(a) + \text{sign}(b)] \min(|a|, |b|). \quad (22)$$

The same reconstruction procedure is applied independently in both the x - and y -directions.

Rusanov Numerical Flux After reconstructing the left and right interface states, the physical fluxes are evaluated and the numerical flux at each interface is computed using the Rusanov (Local Lax–Friedrichs) flux

$$F_{i+\frac{1}{2}} = \frac{1}{2} (F_L + F_R) - \frac{1}{2} \alpha (U_R - U_L), \quad (23)$$

where

$$\alpha = \max(|u_L| + c_L, |u_R| + c_R), \quad (24)$$

denotes the maximum local wave speed, with

$$c = \sqrt{gh}. \quad (25)$$

Similarly, the numerical flux in the y -direction is evaluated as

$$G_{j+\frac{1}{2}} = \frac{1}{2} (G_B + G_T) - \frac{1}{2} \alpha_y (U_T - U_B), \quad (26)$$

where

$$\alpha_y = \max(|v_B| + c_B, |v_T| + c_T). \quad (27)$$

Semi-discrete Finite Volume Formulation After computing the numerical fluxes, the governing shallow water equations are transformed into the following semi-discrete finite volume form

$$\frac{dU_{i,j}}{dt} = -\frac{F_{i+\frac{1}{2},j} - F_{i-\frac{1}{2},j}}{\Delta x} - \frac{G_{i,j+\frac{1}{2}} - G_{i,j-\frac{1}{2}}}{\Delta y} + S_{i,j}. \quad (28)$$

The spatial discretization operator is denoted by

$$R(U) = -\frac{F_{i+\frac{1}{2},j} - F_{i-\frac{1}{2},j}}{\Delta x} - \frac{G_{i,j+\frac{1}{2}} - G_{i,j-\frac{1}{2}}}{\Delta y} + S_{i,j}. \quad (29)$$

SSP–RK3 Time Integration The semi-discrete finite volume system is advanced in time using the third-order Strong Stability Preserving Runge–Kutta (SSP–RK3) scheme.

Stage 1

$$U^{(1)} = U^n + \Delta t R(U^n), \quad (30)$$

Stage 2

$$U^{(2)} = \frac{3}{4}U^n + \frac{1}{4} [U^{(1)} + \Delta t R(U^{(1)})], \quad (31)$$

Stage 3

$$U^{n+1} = \frac{1}{3}U^n + \frac{2}{3} [U^{(2)} + \Delta t R(U^{(2)})]. \quad (32)$$

The MUSCL scheme is a second-order finite-volume reconstruction method in which piecewise

linear states are constructed from cell averages using a slope limiter. In this work, the Minmod limiter is employed to suppress spurious oscillations near discontinuities, while intercell fluxes are evaluated using the Rusanov (local Lax–Friedrichs) flux. The resulting semi-discrete system is advanced in time using the SSP–RK3 method, with the spatial residual recomputed at each Runge–Kutta stage.

Boundary Conditions: Reflective boundary conditions are imposed along all domain boundaries. The water depth is extrapolated using zero-gradient conditions, while the normal momentum component is reflected and the tangential momentum component is extrapolated with zero-gradient conditions.

2.3 Initial Conditions for Dam-Break Scenarios

In this study, six distinct initial water height profiles are considered to evaluate the performance of the numerical schemes. These profiles represent different dam-break configurations, ranging from simple discontinuous jumps to smooth Gaussian and geometric shapes. In all cases, the dam is assumed to be instantaneously removed at $t = 0$, initiating wave propagation. The initial velocity is set to zero throughout the computational domain, i.e.,

$$u(x, y, 0) = 0, \quad v(x, y, 0) = 0. \quad (33)$$

The computational domain is taken as a square region of size

$$L_x \times L_y = 100 \times 100 \text{ m}^2. \quad (34)$$

The bed topography is prescribed as a Gaussian hump:

$$Z(x, y) = 2.0 \exp \left(-\frac{(x - 50)^2 + (y - 50)^2}{200} \right). \quad (35)$$

The gravitational acceleration is set to $g = 2 \text{ m/s}^2$, while the Manning friction coefficient is taken as $n = 0$.

2.3.1 Variant 1: Step (Simple) Dam-Break

This configuration represents the classical dam-break problem extended to two dimensions. The initial water height is defined as

$$h(x, y, 0) = \begin{cases} h_l = 10.0 \text{ m}, & \text{if } x \leq 50 \text{ m}, \\ h_r = 1.0 \text{ m}, & \text{if } x > 50 \text{ m}. \end{cases} \quad (36)$$

This setup creates a sharp discontinuity at $x = 50$ m, producing a shock wave propagating to the right and a rarefaction wave propagating to the left after dam removal.

2.3.2 Variant 2: Rectangular Dam-Break

In this case, a rectangular region of high water is initially confined within the computational domain. The initial condition is given by

$$h(x, y, 0) = \begin{cases} h_{\text{in}} = 5.0 \text{ m}, & \text{if } 30 \leq x \leq 70 \text{ m and } 30 \leq y \leq 70 \text{ m}, \\ h_{\text{out}} = 0.2 \text{ m}, & \text{otherwise.} \end{cases} \quad (37)$$

This configuration models the sudden release of water from a rectangular reservoir, leading to complex wave interactions at the corners and edges.

2.3.3 Variant 3: Circular Dam-Break

The circular dam configuration consists of a cylindrical water column centered within the domain. The initial water height is prescribed as

$$h(x, y, 0) = \begin{cases} h_{\text{in}} = 10.0 \text{ m}, & \text{if } \sqrt{(x - 50)^2 + (y - 50)^2} \leq R, \\ h_{\text{out}} = 1.0 \text{ m}, & \text{otherwise,} \end{cases} \quad (38)$$

where $R = 20$ m denotes the radius of the circular dam.

Following dam removal, a radially symmetric wave propagates outward, providing an effective benchmark for assessing the ability of numerical schemes to capture circular shock fronts.

2.3.4 Variant 4: Gaussian Dam-Break

Unlike the previous discontinuous profiles, the Gaussian initial condition represents a smooth and continuous water surface. The initial water height is defined as

$$h(x, y, 0) = h_{\text{peak}} \exp \left(-\frac{(x - x_c)^2 + (y - y_c)^2}{2\sigma^2} \right), \quad (39)$$

where

- $h_{\text{peak}} = 10.0$ m is the maximum water height,
- $(x_c, y_c) = (50.0, 50.0)$ m denotes the center of the domain,
- $\sigma = 10.0$ m is the standard deviation controlling the spread of the Gaussian profile.

This smooth profile allows the evaluation of numerical schemes in the absence of sharp gradients and tests their ability to preserve the shape and amplitude of smooth waves.

2.3.5 Variant 5: Parabolic Dam-Break

The parabolic dam configuration represents a curved dam structure. The upper and lower boundaries of the parabolic region are defined by

$$y_{\text{top}}(x) = y_c + w - a(x - x_c)^2, \quad (40)$$

$$y_{\text{bottom}}(x) = y_{\text{top}}(x) - t, \quad (41)$$

where the initial water height is prescribed as

$$h(x, y, 0) = \begin{cases} h_{\text{in}} = 10.0 \text{ m}, & \text{if } y_{\text{bottom}}(x) \leq y \leq y_{\text{top}}(x), \\ h_{\text{out}} = 1.0 \text{ m}, & \text{otherwise.} \end{cases} \quad (42)$$

The parameters used are

$$(x_c, y_c) = (50.0, 50.0) \text{ m}, \quad a = 0.03, \quad w = 15.0 \text{ m}, \quad t = 8.0 \text{ m}. \quad (43)$$

This configuration tests the ability of numerical schemes to handle curved boundaries and non-rectangular geometries.

2.3.6 Variant 6: Triangular (Equilateral) Dam-Break

The initial water height profile is defined by an equilateral triangle centered at

$$(x_c, y_c) = (50.0, 50.0) \text{ m}, \quad (44)$$

with side length

$$s = 40.0 \text{ m}. \quad (45)$$

The height of the triangle is given by

$$h_{\text{tri}} = \frac{\sqrt{3}}{2} s = 34.64 \text{ m}. \quad (46)$$

The three vertices of the triangular region are located at

$$\text{Apex (Top):} \quad (x_c, y_c + \frac{h_{\text{tri}}}{3}) = (50.0, 61.55), \quad (47)$$

$$\text{Left Base Vertex:} \quad (x_c - \frac{s}{2}, y_c - \frac{2h_{\text{tri}}}{3}) = (30.0, 26.91), \quad (48)$$

$$\text{Right Base Vertex:} \quad (x_c + \frac{s}{2}, y_c - \frac{2h_{\text{tri}}}{3}) = (70.0, 26.91). \quad (49)$$

The initial water height is prescribed as

$$h(x, y, 0) = \begin{cases} h_{\text{in}} = 10.0 \text{ m}, & \text{if } (x, y) \text{ lies inside the triangular region,} \\ h_{\text{out}} = 1.0 \text{ m}, & \text{otherwise.} \end{cases} \quad (50)$$

The triangular region is defined by the following linear inequalities:

$$y \geq y_{\text{base}}, \quad (51)$$

$$y \leq y_c - \sqrt{3}(x - x_c) + \frac{h_{\text{tri}}}{3}, \quad (52)$$

$$y \leq y_c + \sqrt{3}(x - x_c) + \frac{h_{\text{tri}}}{3}, \quad (53)$$

where

$$y_{\text{base}} = y_c - \frac{2h_{\text{tri}}}{3}. \quad (54)$$

These inequalities ensure that a point lies above the base and below both inclined edges, thereby defining the equilateral triangular water body.

Table 1: Summary of Initial Conditions Considered in This Study

Variant	Name	Type	Inner Height	Outer Height	Special Parameters
1	Step	Discontinuous	10.0 m	1.0 m	$x \leq 50$
2	Rectangular	Discontinuous	5.0 m	0.2 m	$30 \leq x, y \leq 70$
3	Circular	Discontinuous	10.0 m	1.0 m	$R = 20$ m
4	Gaussian	Smooth	10.0 m (peak)	N/A	$\sigma = 10$ m
5	Parabolic	Curved	10.0 m	1.0 m	$a = 0.03, w = 15, t = 8$
6	Triangular	Equilateral	10.0 m	1.0 m	$s = 40$ m

2.3.7 Numerical Setup

For all six dam-break scenarios, the following numerical parameters are employed:

- Computational domain: $[0, 100] \times [0, 100]$ m².
- Grid resolution: $N_x = N_y = 501$.
- Spatial step size: $\Delta x = \Delta y = 0.2$ m.

- Total simulation time: $T = 2.0$ s.
- Gravitational acceleration: $g = 2$ m/s².
- Boundary conditions: Reflective boundary conditions are imposed along all boundaries.
- Artificial viscosity: Additional artificial viscosity is employed in the Lax–Wendroff scheme to suppress spurious oscillations.

The numerical solutions obtained for all six configurations serve as benchmark datasets for evaluating the performance of the numerical schemes presented in the subsequent sections.

2.4 Scientific Machine Learning Models

We evaluate two families of neural solvers against the classical reference solutions, namely strong-form physics-informed neural networks (PINNs) and a finite-volume-informed PINN (FVM–PINN). All models are trained per scenario, so that one network approximates the space-time solution of a single dam-break configuration.

2.4.1 Physics-Informed Neural Networks

The PINN parameterizes the solution with a fully connected network $\mathcal{N}_\theta : (x, y, t) \mapsto \mathbb{R}^3$, with inputs normalized to $[-1, 1]^3$ and an optional random Fourier feature embedding $z \mapsto [\sin(2\pi Bz), \cos(2\pi Bz)]$ to mitigate spectral bias [20]. Two output formulations are compared. In the primitive formulation the network predicts (h, u, v) directly, whereas in the conservative formulation it predicts the conserved variables (h, hu, hv) , matching the form in which the weak solution is defined. In both cases the residuals of eq. (1) (including the bed-slope source term of eq. (4)) are evaluated pointwise by automatic differentiation at collocation points sampled in the space-time domain, and the network is trained on the composite loss

$$\mathcal{L}(\theta) = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}} + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}, \quad (55)$$

where $\mathcal{L}_{\text{pde}}$ is the mean squared PDE residual, $\mathcal{L}_{\text{ic}}$ anchors the prediction at $t = 0$ to the initial condition, and the optional $\mathcal{L}_{\text{data}}$ penalizes the misfit against sparse observations taken from the reference solution (used only in the data-guidance ablation). The network has six hidden layers of width 128 with tanh activations. At each iteration the residual is evaluated at 20,000 collocation points drawn uniformly over the space-time domain, and the initial condition is imposed at every cell centre of the evaluation grid. The loss weights are $\lambda_{\text{pde}} = 1$, $\lambda_{\text{ic}} = 10$, and $\lambda_{\text{data}} = 5$.

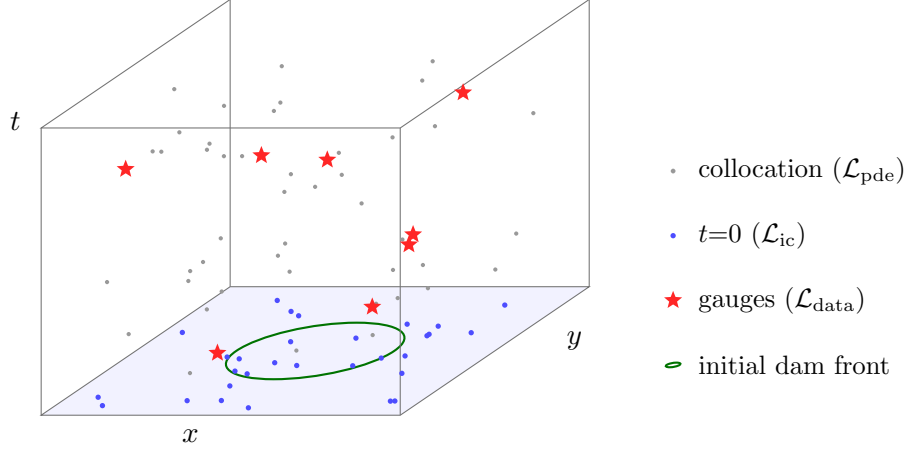


Figure 1: Space-time sampling of the training points over the domain and time horizon.

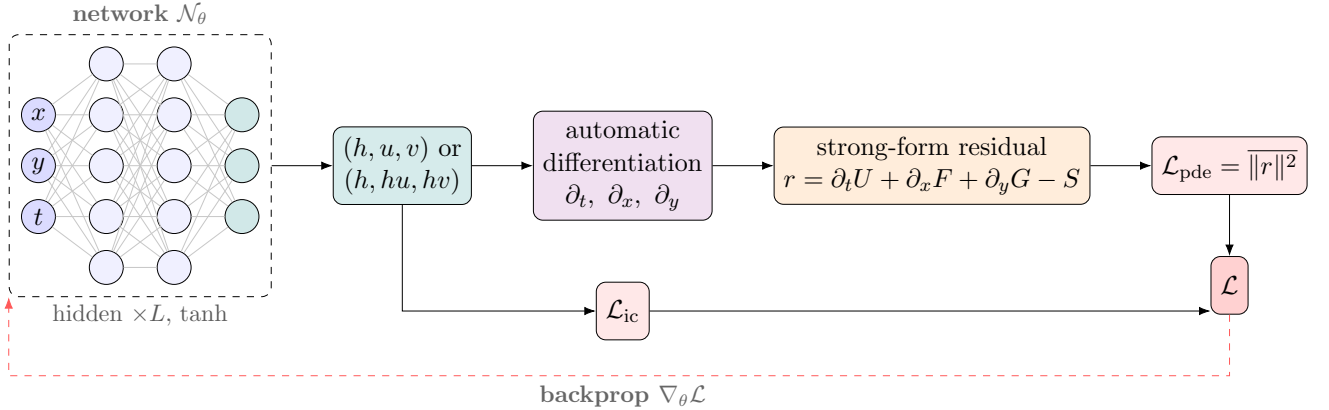


Figure 2: Strong-form PINN: the network outputs are differentiated to form the SWE residual, which with the initial-condition term makes the loss.

2.4.2 Finite-Volume-Informed PINN

Strong-form residuals are ill-defined at the shocks that dominate dam-break flows [18, 19], and strong-form PINNs do not enforce discrete conservation [21]. The FVM-PINN therefore replaces

the pointwise residual with a *discrete-consistency* residual evaluated through a differentiable finite-volume solver. The network predicts a free-surface perturbation ξ_θ together with the two velocity components, and the water depth is recovered as

$$h(x, y, t) = \text{softplus}(\xi_\theta(x, y, t) + h_s(x, y)), \quad (56)$$

where h_s is the initial depth field. Writing the depth this way keeps it strictly positive by construction and lets the network learn a perturbation around the initial state. The momentum components are formed as $hu = h u_\theta$ and $hv = h v_\theta$, with the predicted velocities passed through a bounded activation. Momentum therefore vanishes wherever the depth goes to zero, which keeps the solver step finite on dry beds and removes the velocity blow-up that a direct momentum output would otherwise produce. Let $\mathcal{S}_{\Delta t}$ denote one step of the differentiable finite-volume solver (HLLC flux with MUSCL reconstruction and SSP–RK time integration, as in Sect. 2.2, implemented end-to-end differentiably). With $U_\theta(t_k)$ the network state evaluated on the full grid at time t_k , the discrete residual over the interval $[t_k, t_{k+1}]$ is

$$\mathcal{L}_{\text{fvm}}^{(k)} = \|U_\theta(t_{k+1}) - \mathcal{S}_{\Delta t}(U_\theta(t_k))\|_2^2. \quad (57)$$

At each training iteration a single interval index k is sampled at random (stochastic single-interval residual), which keeps the per-iteration cost at one solver step while covering the full time horizon in expectation. The total loss combines eq. (57) with an initial-condition anchor and, in the data-guidance ablation, a sparse-data anchor following the finding of [21] that physics-only finite-volume losses admit spurious low-momentum minima.

Figure 3 shows the complete training loop: the shared network predicts $(\xi, \tilde{u}, \tilde{v})$, the reparametrisation of eq. (56) recovers the conserved state on the solver grid, and one differentiable solver step (MUSCL, Audusse, HLLC, SSP–RK2) advances $U_\theta(t_k)$ so that the discrepancy with $U_\theta(t_{k+1})$ gives the residual. Gradients of the total loss backpropagate through the solver step to θ .

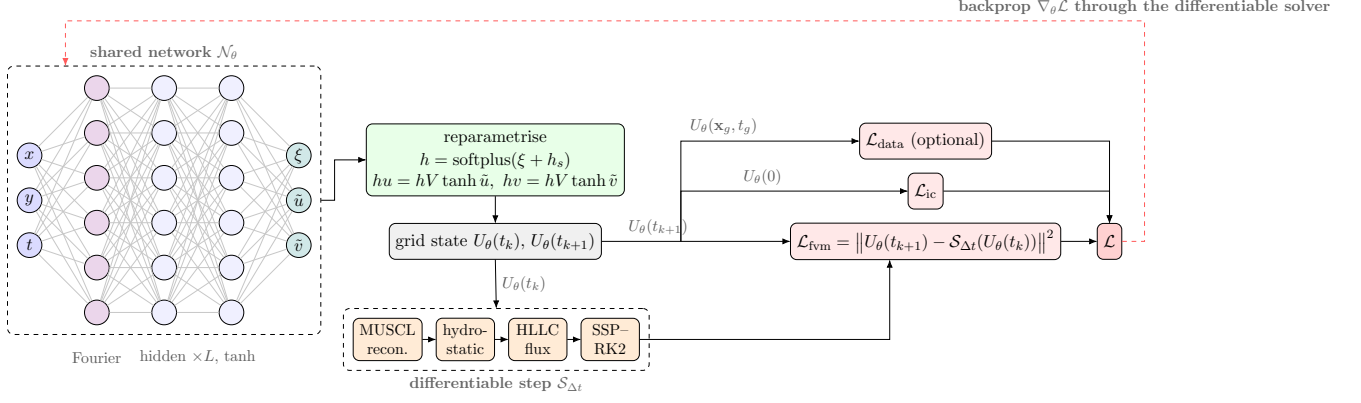


Figure 3: FVM-informed PINN architecture and training loop.

2.4.3 Training Protocol and Evaluation Metrics

Both families use tanh networks. The strong-form PINNs have six hidden layers of width 128, and the FVM-PINN has five hidden layers of width 128 preceded by a 32-frequency Fourier-feature embedding. All models are trained with the Adam optimizer at a constant learning rate of 10^{-3} for 12,000 iterations with gradient-norm clipping, and each configuration is repeated over three random seeds with metrics reported as mean $\pm$ standard deviation. The differentiable solver step inside the FVM-PINN loss is evaluated in double precision. Training was carried out on an NVIDIA Quadro RTX 8000 GPU.

The neural models are evaluated across the initial-condition variants of Sect. 2.3, under the same domain, bed topography, and boundary conventions as the classical schemes, so that the neural results extend the classical comparison matrix along its initial-condition axis. A dry-bed circular case, in which the downstream depth is set to zero, is added to probe the low-momentum collapse of the physics-only finite-volume loss and the effect of sparse-data guidance. For each case a fine-grid solution computed with the differentiable HLLC solver of Sect. 2.2 serves as the reference; it is resampled onto the evaluation grid and compared with the network prediction at the output times $t \in \{0.5, 1.0, 1.5, 2.0\}$ s using:

- the discrete L^1 error of the water depth h , integrated over cell areas and reported as the mean over the three seeds;
- relative mass drift $\max_t |V(t)/V(0) - 1|$, where $V(t)$ is the total water volume, quantifying

conservation.

3 Results and Discussion

We present the numerical results for the six dam-break examples that we consider in this work, obtained using the Lax–Wendroff (LW), Harten–Lax–van Leer (HLL), and MUSCL–Rusanov (MUSCL–RS) schemes. The main goal is to study and evaluate the performance of these numerical methods with respect to accuracy, stability, shock capturing and capability of modeling shallow water flows over non-flat bed topography.

Numerical simulations are carried out on a 2D computational domain that includes a Gaussian bed elevation as a source term. The temporal evolution of the free-surface flow is analysed at selected times: $t = 0.0, 0.5, 1.0, 1.5$ and 2.0 s. The considered benchmark problems are discontinuous, smooth, and geometrically complex initial conditions which are the Step, Rectangular, Circular, Gaussian, Parabolic, and Triangular dam-break problems.

Overall, all three numerical schemes can reproduce the overall hydrodynamic properties of the dam-break flows. But the solutions that are computed do show significant differences. The LW scheme typically gives better resolution of discontinuities and can have oscillatory behavior in the vicinity of steep gradients. The HLL scheme is stable and monotone, but it introduces additional numerical diffusion resulting in more smooth shock profiles when compared with other schemes. The MUSCL–Rusanov scheme, however, is able to retain sharp wave fronts while suppressing non-physical oscillations effectively in all of the tests, thus showing best overall performance in the schemes tested herein.

3.1 Variant 1: Step Dam-Break

The numerical results of the step dam break problem are given in Fig. 4. At first, a discontinuity in water depth is prescribed at the middle of the calculation domain. The shock wave will travel downstream and a rarefaction wave will travel upstream after the dam is removed in an instant.

During the simulation, the propagating wave encounters the underlying Gaussian bed topogra-

phy which produces noticeable changes in the free surface profile. All three numerical schemes are able to capture the main flow features. But there's a big disparity in their ability to capture shocks. The LW scheme is able to capture discontinuities well but has oscillatory behaviour close to steep gradients. The HLL scheme gives very stable and monotonic solutions, but with more numerical diffusion, which leads to shock profiles that are comparatively smooth. The MUSCL–Rusanov scheme on the other hand is able to generate the flow evolution without producing non-physical oscillations and with sharp wave fronts, giving the most accurate representation of the flow evolution.

Stepped_Comparison of LW, HLL and MUSCL-Rusanov Schemes

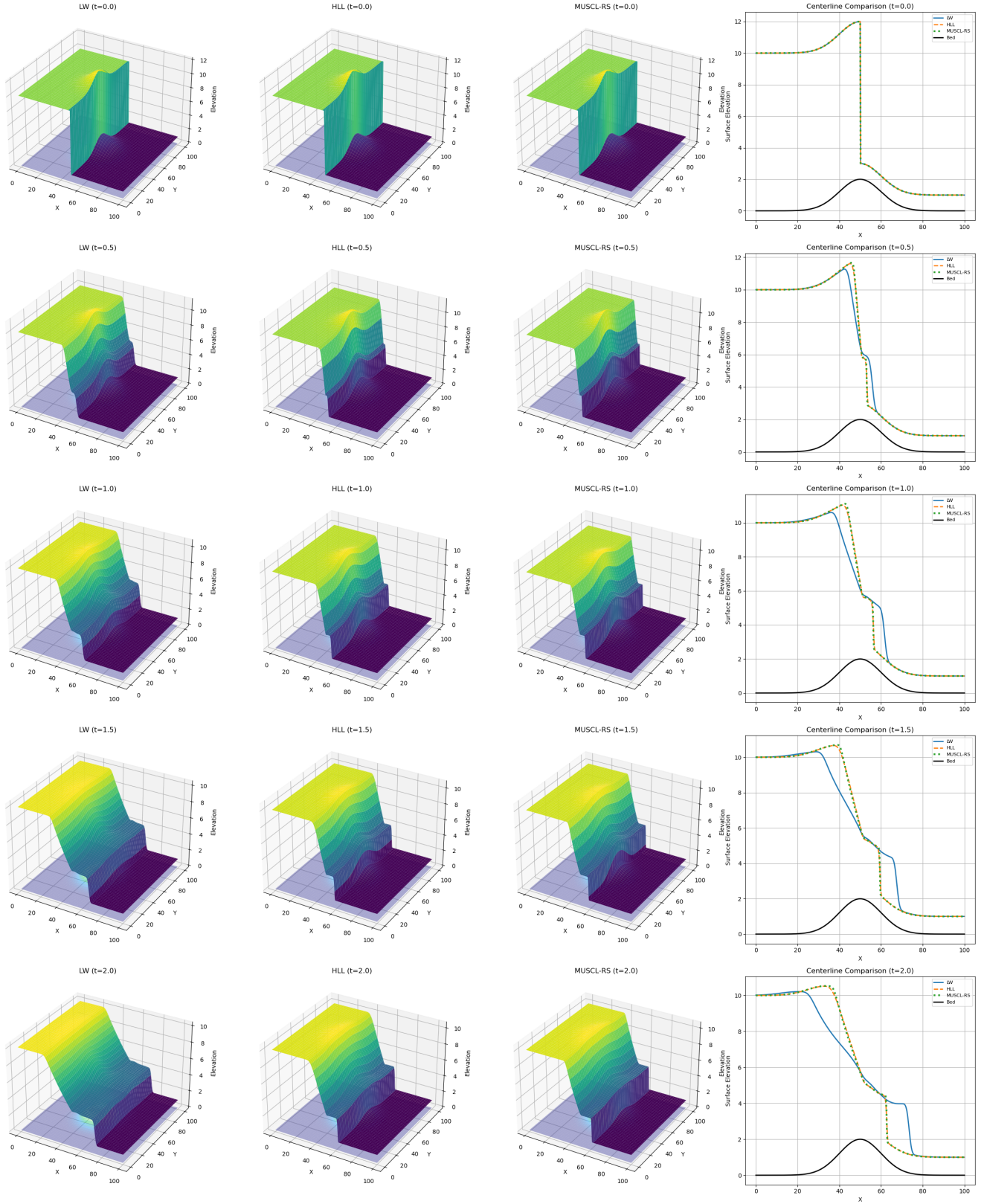


Figure 4: Comparison of the LW, HLL, and MUSCL–Rusanov schemes for the step dam-break problem at different time instances.

3.2 *Variant 2: Rectangular Dam-Break*

The numerical solutions for the rectangular dam-break configuration are shown in figure 5. In this situation, the water is first trapped in a rectangular area in the middle of the computational area. After the removal of the dam, the water flows out in all directions, resulting in multidirectional wave propagation and strong interactions in the corners of the reservoir.

Because of the interaction of waves at the edges and corners, these waves create complex structures in the rectangular geometry. All numerical schemes capture the overall spreading pattern, but there are discernible differences in the spreading near the discontinuities. The LW scheme is able to reproduce the growing fronts properly, while the regions with sharp gradients have oscillations. The HLL scheme produces stable and smooth solutions, but with a higher numerical diffusion. The MUSCL-Rusanov scheme provides a good solution and is able to keep the interface sharp and be numerically stable while giving a correct solution for the corner interactions.

Rectangular Dam-Break Comparison with Bed Topography

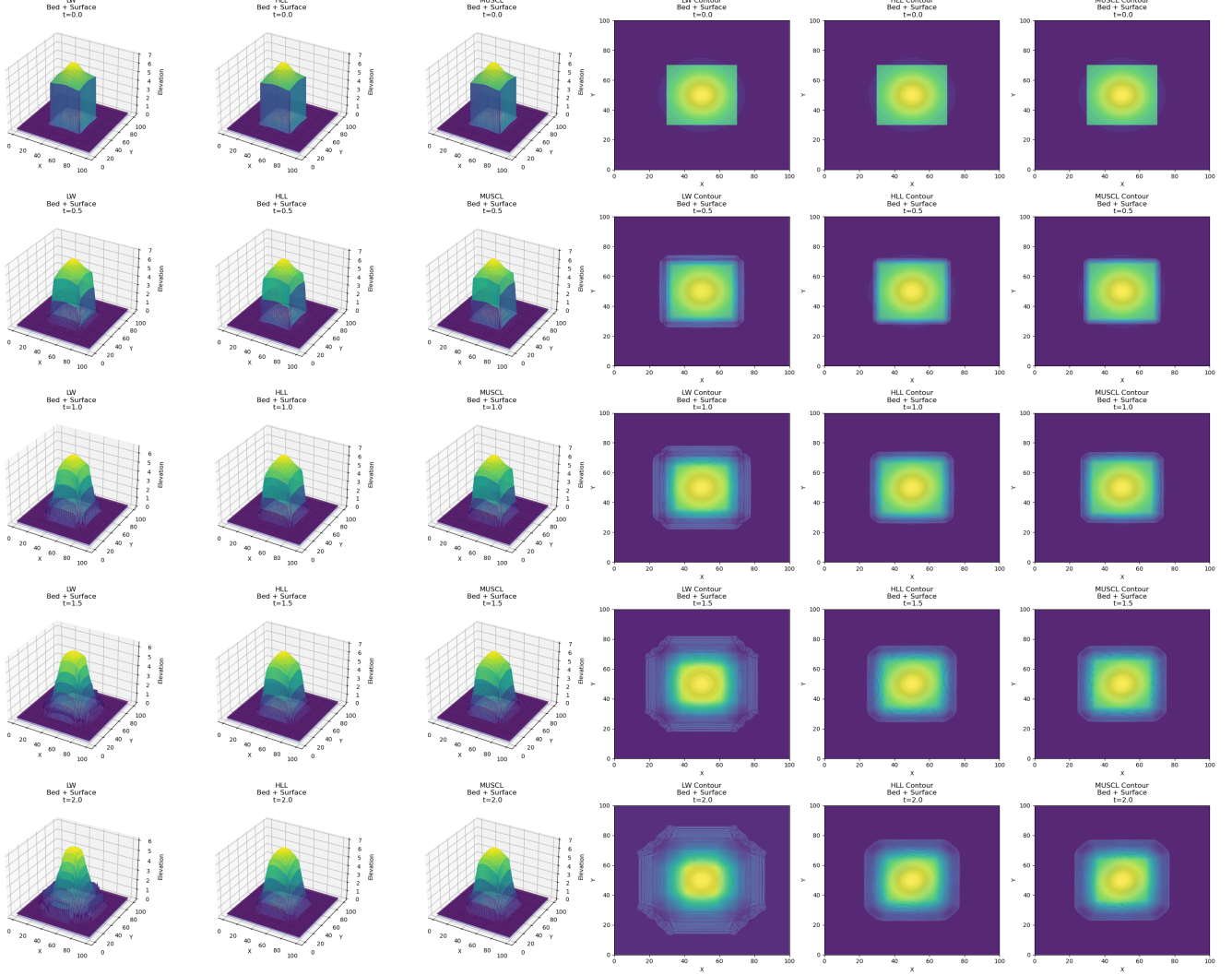


Figure 5: Comparison of the LW, HLL, and MUSCL–Rusanov schemes for the Rectangular dam-break problem at different time instances.

3.3 Variant 3: Circular Dam-Break

The numeric solution results for the dam break problem with circular dams are displayed in Fig. 6. The first water column that is confined to a cylinder is used as a source to set up radially symmetric waves that propagate outwards from the centre of the domain after the dam has been suddenly removed.

The overall radial symmetry of the flow is maintained during the simulation with all the numerical schemes. However, there are variations in the representation of the growing shock front. The LW scheme is able to accurately represent the propagating wave, but has some oscillations in

the vicinity of the shock region. The HLL scheme is diffusive and hence gives smoother solutions. This is in contrast with the MUSCL–Rusanov scheme which is able to capture the shock front as well as maintain the radial symmetry and avoid the appearance of numerical oscillations. As a result, the MUSCL–Rusanov method is shown to perform better for circular wave propagation.

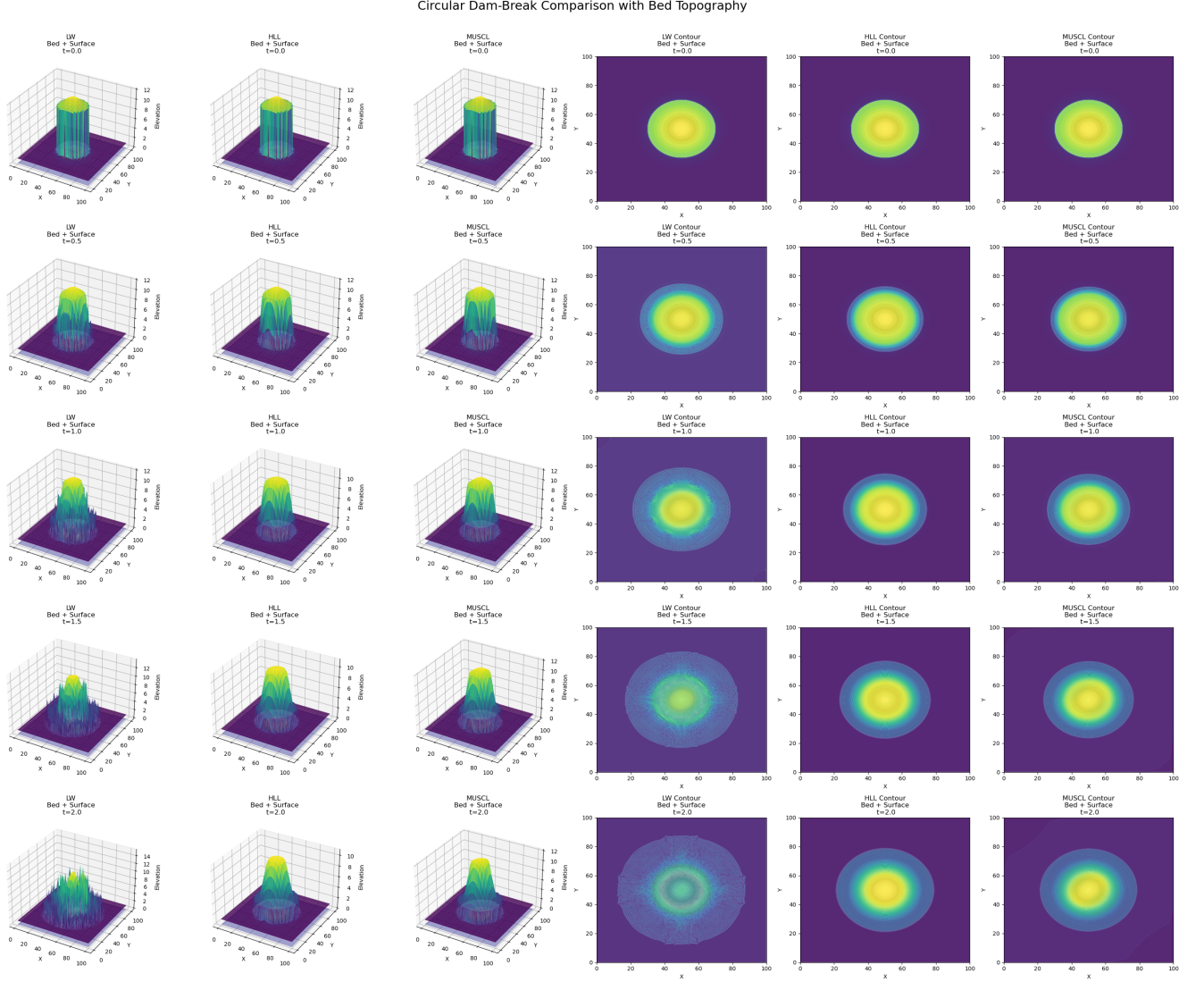


Figure 6: Comparison of the LW, HLL, and MUSCL–Rusanov schemes for the Circular dam-break problem at different time instances.

3.4 Variant 4: Gaussian Dam-Break

The time evolution of the Gaussian dam-break configuration is shown in Figure 7. In contrast to the previous discontinuous cases, the Gaussian initial condition is a smooth free surface profile, with a continuous water depth distribution.

There are no initial sharp discontinuities and all three numerical schemes produce good results and retain the smoothness of the propagating wave. The LW scheme is highly accurate but can give some dispersive effects in a long time integration. The HLL scheme shows no change in the stability of the scheme during the simulation, but there is a little bit of numerical diffusion. The MUSCL–Rusanov scheme is able to preserve the amplitude and the shape of the Gaussian profile better, and thus the smooth flow evolution is best represented by this scheme.

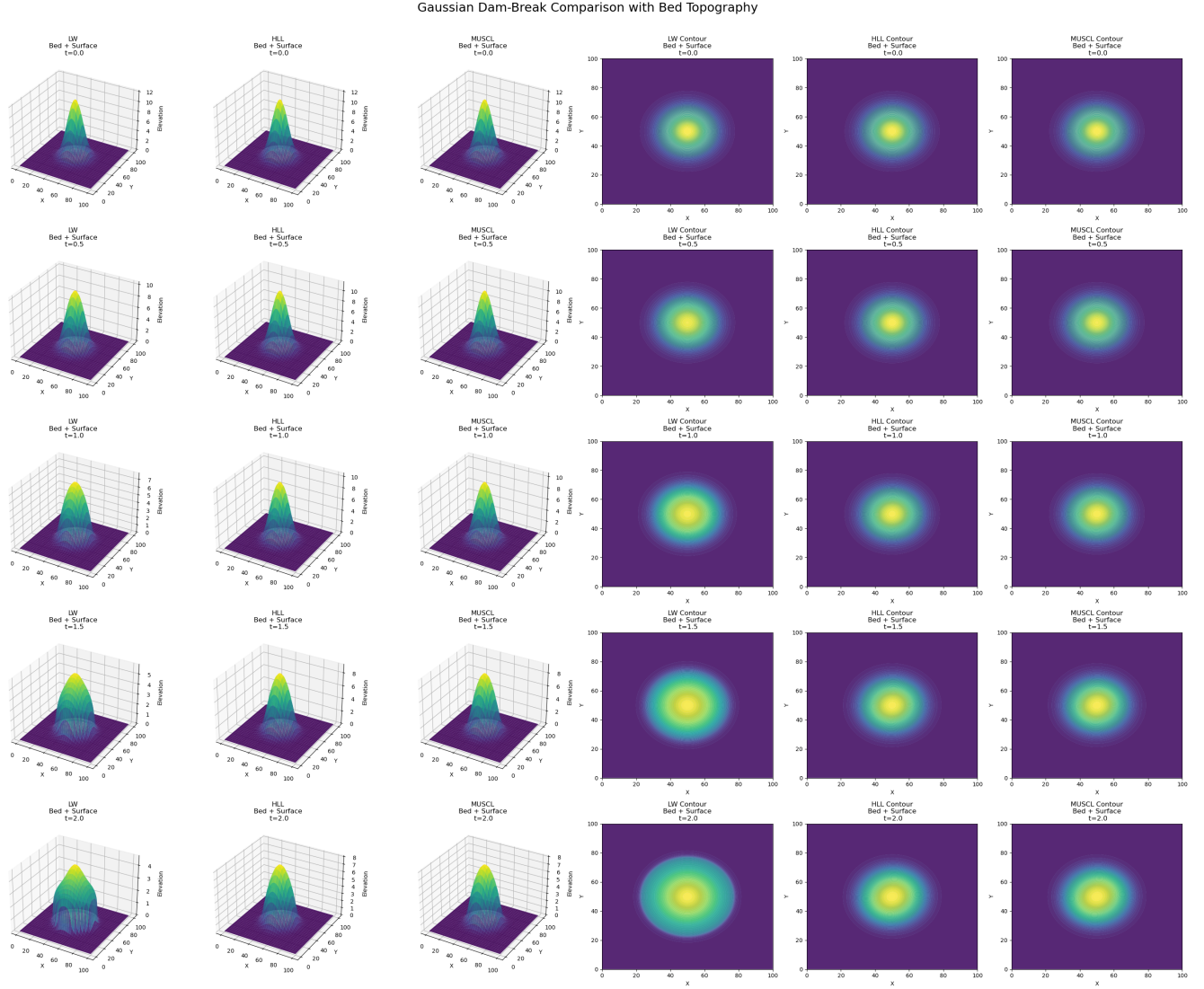


Figure 7: Comparison of the LW, HLL, and MUSCL–Rusanov schemes for the Gaussian dam-break problem at different time instances.

3.5 *Variant 5: Parabolic Dam-Break*

In figure 8 numerical solution to the parabolic dam-break problem is shown. The underlying bed topography has a complex relationship with the water profile, the propagation patterns are not uniform, and are curved in the first water profile.

This curved boundary creates asymmetric patterns in the flow, which also results in changes in the wave speed at certain points. All schemes are able to reproduce the dominant hydrodynamic features, but the LW scheme shows oscillatory behaviour in the region of the steep gradients. The HLL scheme is stable, but the propagating wave fronts tend to smear, because of numerical diffusion. The MUSCL–Rusanov scheme, on the other hand, preserves the curved interfaces correctly and has excellent stability and stability minimizing spurious oscillations. The results presented here validate the robustness of the MUSCL–Rusanov scheme on the geometrically complex flow configurations.

Parabola Dam-Break Comparison with Bed Topography

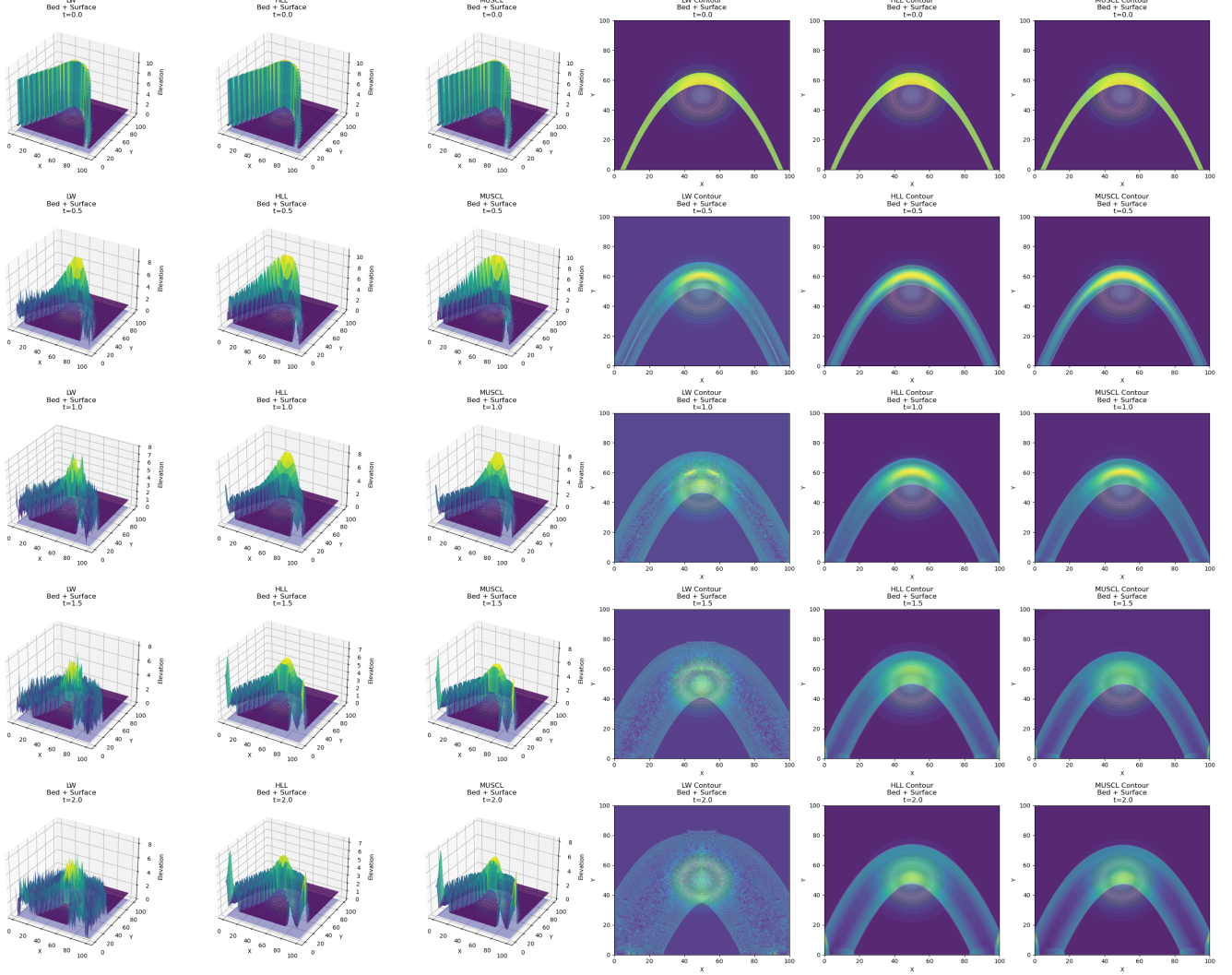


Figure 8: Comparison of the LW, HLL, and MUSCL–Rusanov schemes for the Parabola dam-break problem at different time instances.

3.6 Variant 6: Triangular Dam-Break

The numerical results for the triangular dam-break configuration are shown in figure 9. This benchmark problem is complicated by sharp vertices and tilted boundaries, which produce complex interactions of the waves and strong shock reflections.

The propagating waves that come from the triangular vertices interact with each other and form complex flow structures throughout the computational domain. The LW scheme clearly reproduces these interactions, but there are discontinuities and noticeable oscillations near the vertices. This HLL scheme successfully suppresses oscillations but results in relatively broad wave

fronts. The MUSCL–Rusanov scheme is accurate in solving the corner interactions and retains the overall triangularity of the solution with the stability of the numerical solution. Overall, the MUSCL–Rusanov scheme is the most accurate and stable for the triangular dam-break problem.

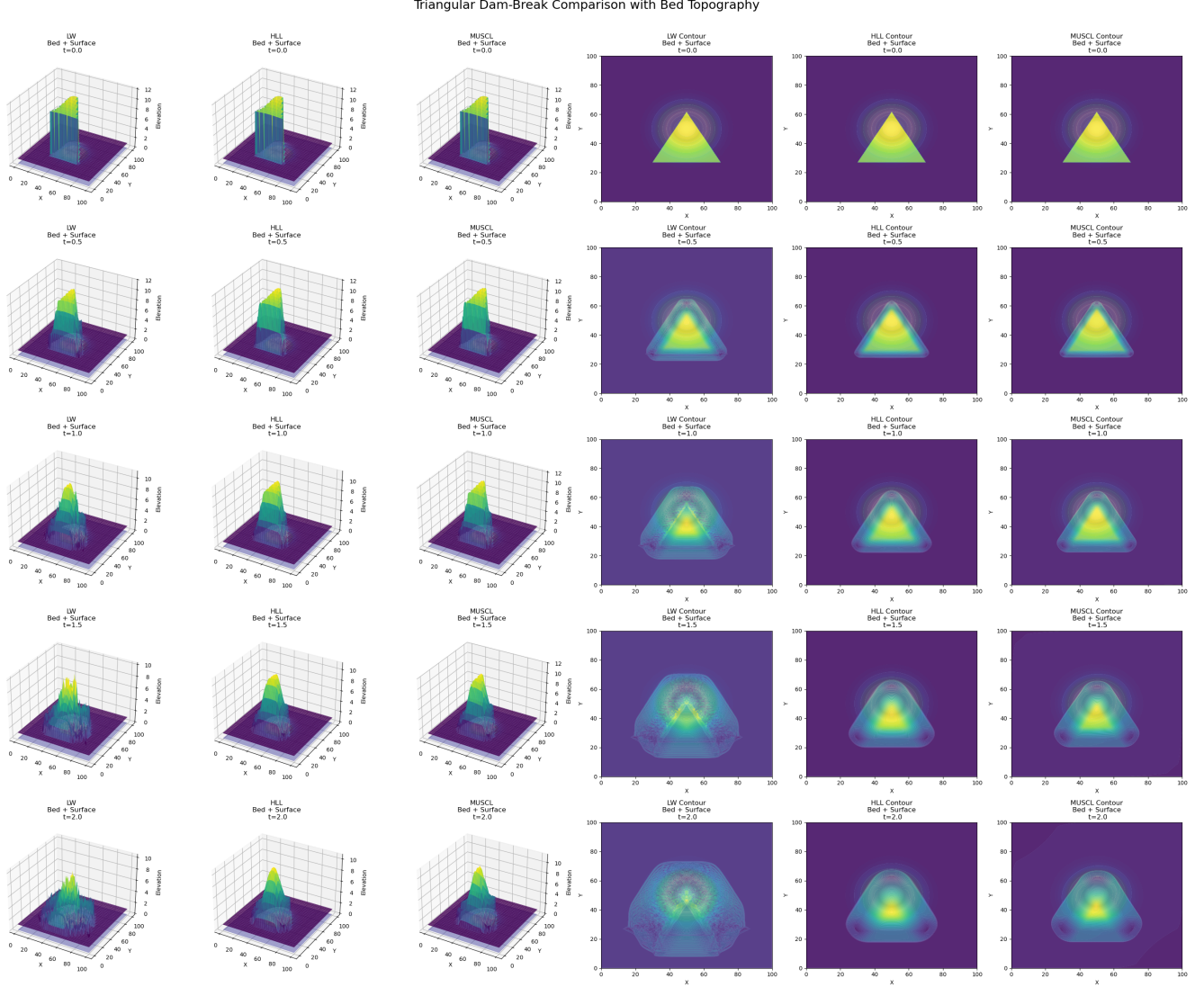


Figure 9: Comparison of the LW, HLL, and MUSCL–Rusanov schemes for the Triangular dam-break problem at different time instances.

3.7 SciML Comparison Across Initial Dam-Breaking Conditions

This section extends the comparison of Sects. 3.1–3.6 with the neural formulations, evaluated over all six initial-condition variants under the same domain, bed topography, and boundary conventions. For the six benchmark variants the reference is the MUSCL–Rusanov depth solution

of Sects. 3.1–3.6 – the most accurate of the classical schemes considered here – bilinearly resampled from its 501×501 grid onto the 128×128 evaluation grid; errors are the depth $L^1(h)$ at the final time $t = 2$ s, averaged over three seeds. The classical baseline row of Table 2 is the HLL scheme against the same reference (MUSCL–Rusanov itself defines the reference and is therefore omitted), so it reports the accuracy floor spanned by the classical schemes and every neural model is measured against the identical reference. To confirm that the neural training pipeline is consistent with this reference, the differentiable HLLC solver used to build the collocation losses and the ablation references below was reconciled against the classical circular solution in matched configuration and agrees to within 0.17% relative L^1 . Table 2 reports accuracy across the six variants and Table 3 the corresponding relative mass drift, the latter computed from each method’s own predicted volume and hence independent of the reference.

Table 2: Depth error $L^1(h)$ at $t = 2$ s (integrated over cell areas) for each method across the six initial-condition variants, measured against the MUSCL–Rusanov reference resampled to the 128×128 evaluation grid. Neural entries are the mean over three seeds; the classical column is the HLL scheme at the same resolution, giving the classical accuracy floor.

Variant	Classical (HLL)	PINN (prim.)	PINN (cons.)	FVM–PINN
1 Step	273	3000	3230	11000
2 Rectangular	147	790	836	4990
3 Circular	831	1550	2770	13500
4 Gaussian	49	332	601	5170
5 Parabolic	681	2670	3130	13500
6 Triangular	500	1150	2140	8940

Three observations hold across the initial-condition matrix. First, the error tracks the smoothness of the initial data: the smooth Gaussian mound (Variant 4) is by far the easiest case for every method – the classical HLL floor is $L^1(h) = 49$ and the primitive PINN reaches 332, well below the other variants – whereas the thin curved parabolic band (Variant 5) is the hardest, and the discontinuous step, circular, and triangular columns fall in between. Second, among the strong-form networks the primitive formulation is more accurate than the conservative one on every variant except the rectangular column, where the two are comparable; writing the residual directly on

Table 3: Relative mass drift $\max_t |V(t)/V(0) - 1|$ for each method across the six variants. The finite-volume schemes conserve mass to round-off; the neural models do not enforce conservation exactly.

Variant	Classical	PINN (prim.)	PINN (cons.)	FVM-PINN
1 Step	~ 0	0.002	0.019	0.046
2 Rectangular	~ 0	0.013	0.026	0.276
3 Circular	~ 0	0.008	0.038	0.390
4 Gaussian	~ 0	0.003	0.030	0.540
5 Parabolic	~ 0	0.008	0.063	0.460
6 Triangular	~ 0	0.009	0.029	0.280

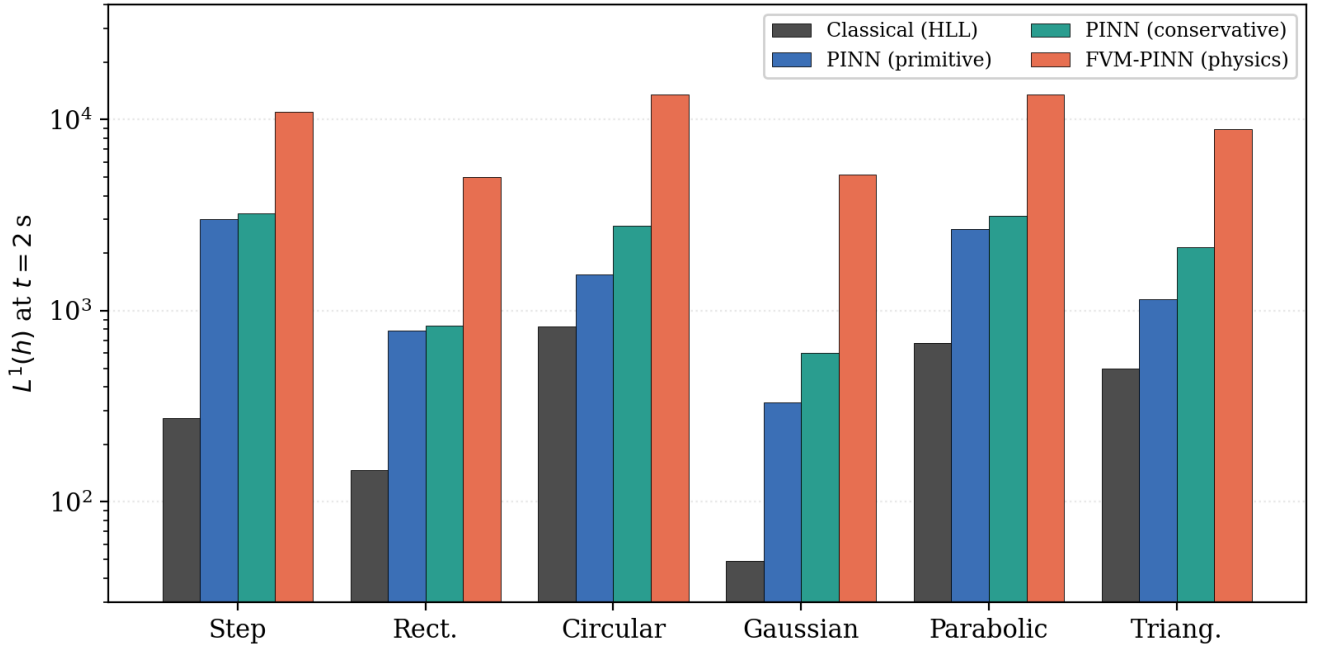


Figure 10: Depth error $L^1(h)$ at $t = 2$ s across the six initial-condition variants (log scale).

the conserved variables did not help in this setting. Third, the physics-only FVM-PINN is the least accurate neural model on every variant and drifts the most (up to 0.5), the low-momentum collapse documented by Liu [21]: the network settles into a slowly varying, nearly static state that is self-consistent under the discrete solver step and matches the initial condition, yet fails to reproduce the true wave propagation. The collapse is most severe precisely where the flow is most dynamic. The dry-bed case and the shock-strength contrast below examine this collapse and the conditions under which it is removed.

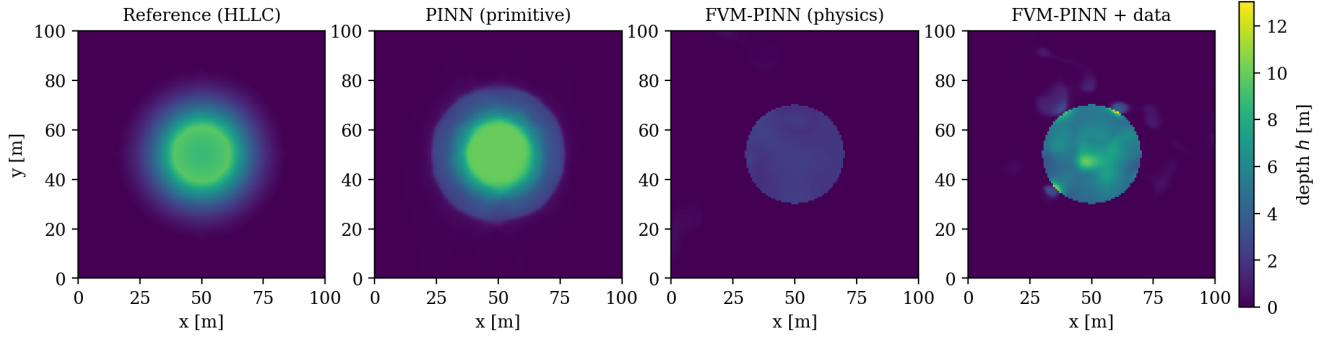


Figure 11: Depth field at $t = 2$ s on the dry-bed circular case: fine-grid HLLC reference and the three neural predictions on the evaluation grid.

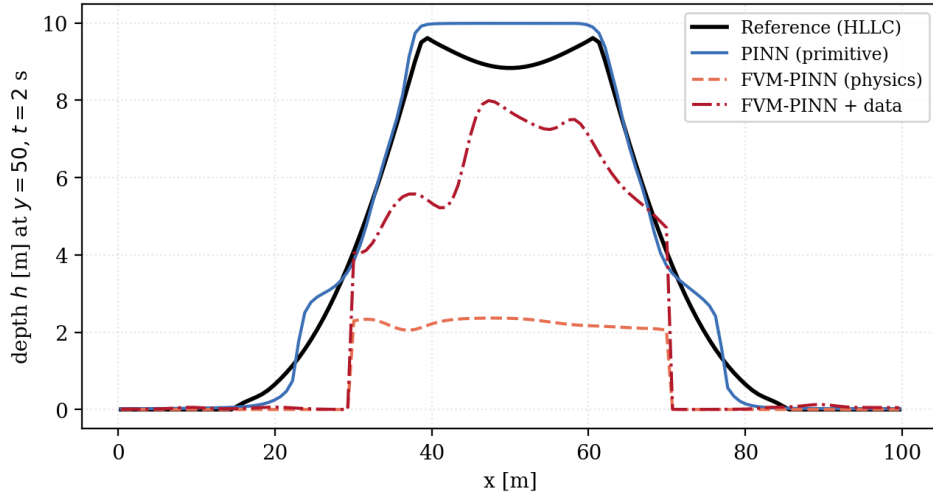


Figure 12: Centreline depth profile ($y = 50$ m, $t = 2$ s) on the dry-bed circular case.

The collapse is visible directly in Figure 11: the physics-only FVM–PINN predicts a nearly static, low-amplitude disk that has scarcely spread from the initial column, whereas the primitive PINN reproduces the radial wave and the classical reference sets the standard. The centreline profile of Figure 12 makes the same point quantitatively: the physics-only prediction sits flat near 2 m across the original column footprint, while adding sparse data lifts the amplitude back towards the reference.

3.7.1 Data-guidance ablation

Adding the sparse-data term of eq. (55) to the FVM–PINN loss steadily removes the collapse. Table 4 reports the dry-bed circular case as the number of gauge points, drawn from the reference solution, increases. Both the depth error and the mass drift fall monotonically, from the physics-

only values towards the strong-form range; with 1024 gauges (about 6% of the grid cells at a single snapshot) the error is roughly a third of the physics-only value and the drift is below 0.08. This is the quantitative form of the observation in [21] that a small amount of data restores the missing momentum content, and it confirms that the collapse is an optimisation pathology of the physics-only objective rather than a failure of the discrete-residual formulation itself.

Table 4: Sparse-data guidance on the dry-bed circular case: depth error and mass drift versus the number of reference gauge points, mean over three seeds.

Gauge points	$L^1(h)$	mass drift
0 (physics only)	9210	0.42
64	6590	0.17
256	5670	0.14
1024	3410	0.08

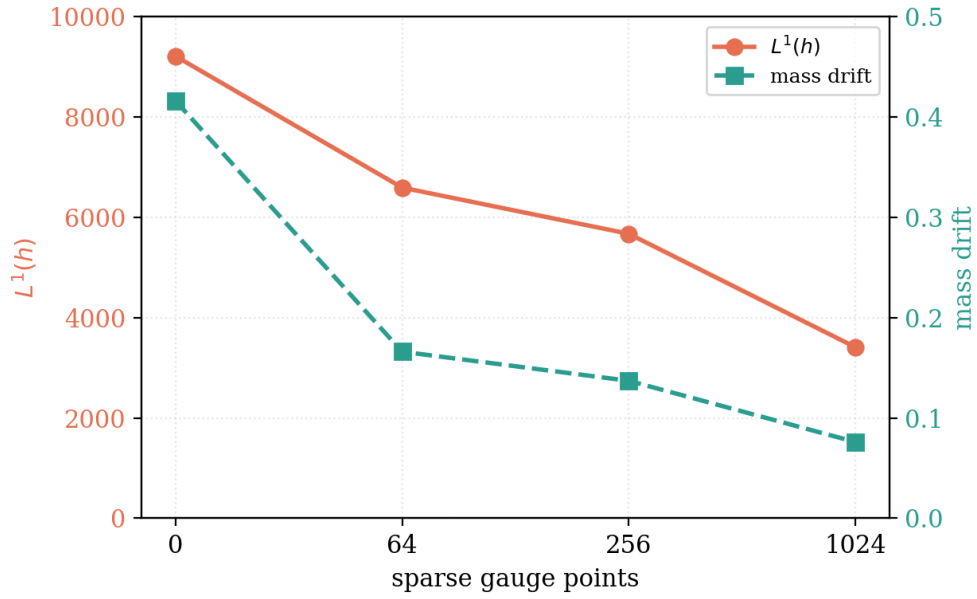


Figure 13: Data-guidance recovery on the dry-bed circular case: depth error and mass drift versus the number of sparse gauge points.

3.7.2 Effect of shock strength

The benchmark configuration of Sect. 2.3 ($g = 2 \text{ m/s}^2$, $t \leq 2 \text{ s}$) produces comparatively slow waves and moderate shocks, a regime in which the strong-form PINN’s characteristic weakness, the

smearing of sharp fronts, is only mildly exercised. To probe whether the ordering depends on shock strength, the circular dry-bed case was repeated in a deliberately stronger-shock configuration ($g = 9.81 \text{ m/s}^2$, a tighter column of radius $R = 11 \text{ m}$ on a flat bed, evaluated at $t = 1.2 \text{ s}$). This departs from the benchmark conventions of Sect. 2.3 precisely to isolate the effect of shock strength; because both regimes are ablations of the dry-bed circular case with no classical reference in the benchmark set, the reference here is a fine-grid HLLC self-convergence solution rather than the MUSCL–Rusanov data used in Table 2. Table 5 contrasts the two regimes.

Table 5: Effect of shock strength on the dry-bed circular case: depth error $L^1(h)$ (mean over three seeds) in the moderate-shock benchmark regime ($g = 2 \text{ m/s}^2$) and a stronger-shock regime ($g = 9.81 \text{ m/s}^2$).

Method	$L^1(h), g = 2$	$L^1(h), g = 9.81$
Classical HLLC	380	79
PINN (primitive)	1680	1120
PINN (conservative)	2040	2410
FVM–PINN (physics only)	9020	3080
FVM–PINN + data (256)	5670	1950

The effect is pronounced. As the shock strengthens, the FVM–PINN’s depth error falls by roughly a factor of three, both with and without data guidance, whereas the strong-form PINNs change comparatively little. The ordering among the neural methods rearranges accordingly: in the stronger-shock regime the data-guided FVM–PINN overtakes the conservative PINN and approaches the primitive one, having been the least accurate neural model in the moderate-shock case. This is consistent with the view that the FVM–PINN’s discrete-residual loss acts as a shock-capturing prior whose benefit grows with the strength of the discontinuities the network must represent, so its structural advantage is realised most clearly on strong dam-break fronts rather than on the slower flows of the present benchmark set.

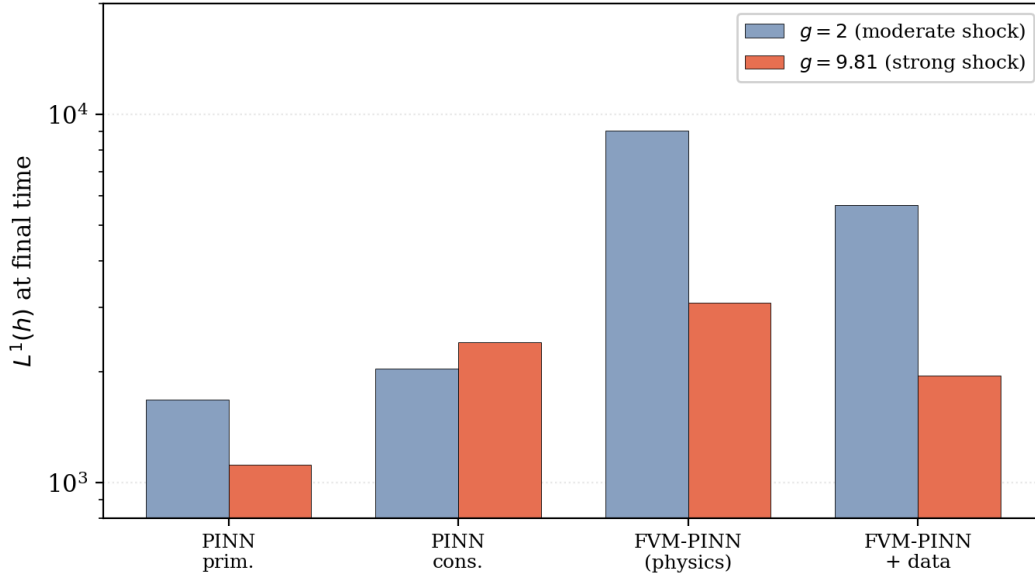


Figure 14: Shock-strength contrast on the dry-bed circular case: depth error at $g = 2$ and $g = 9.81 \text{ m/s}^2$ (log scale).

3.8 Discussion

The comparison places three neural formulations against classical finite-volume schemes on identical dam-break problems, and the results temper the common expectation that embedding discrete structure into the loss automatically yields a better solver.

Classical schemes remain the accuracy-per-cost reference. No neural method approaches the classical finite-volume error at the training budget used here; a single solver run is both faster and more accurate than training any of the networks for one instance. The value proposition of the neural models is therefore not accuracy but the mesh-free, differentiable, space-time representation they provide, together with the amortisation of inference once training is paid for.

Formulation matters more than expected among strong-form PINNs. The primitive (h, u, v) formulation outperformed the conservative (h, hu, hv) one on every case. Although the conservative form is the one in which the weak solution is defined, its residual couples the momentum components multiplicatively through the flux, which appears to make the optimisation harder here without a compensating accuracy gain at these weak shocks.

Discrete structure is not a free lunch. The FVM–PINN inherits discrete conservation and well-balancing *only if* its predicted trajectory actually minimises the discrete residual along the true solution. Trained on physics alone it instead finds a spurious low-momentum trajectory that also satisfies the residual, and collapses. Sparse data removes this failure mode in a controlled, monotone way, but in the moderate-shock regime of the present benchmarks even the data-guided FVM–PINN does not overtake the far simpler primitive PINN. The shock-strength contrast of Sect. 3.7.3 clarifies why: as the shock strengthens the FVM–PINN’s error falls by about a factor of three and the data-guided variant climbs past the conservative PINN, so the discrete-residual structure behaves as a shock-capturing prior whose benefit is realised on strong fronts rather than on the slower flows of these benchmarks. This makes the FVM–PINN’s standing regime-dependent rather than uniformly weak.

Limitations. The neural models are trained per instance and do not generalise across scenarios; cross-scenario generalisation is the province of operator learning [26–28], the natural next step for this differentiable solver. The setting is also idealised ($n = 0$, reflective walls, the benchmark value $g = 2 \text{ m/s}^2$ of Sect. 2.3), and the references are grid-bound fine solutions rather than analytic ones. Finally, the training budget was fixed across methods for fairness rather than tuned per method, so the absolute neural errors should be read as characteristic of a modest budget rather than as best-achievable.

4 Conclusion

We compared classical shock-capturing schemes and physics-informed neural networks on two-dimensional dam-break problems with bed-slope source terms across a range of initial dam-breaking conditions. Among the classical schemes, the MUSCL–Rusanov method gives the best overall balance of shock sharpness and stability across all six variants, the HLL solver is robust but more diffusive, and the Lax–Wendroff scheme resolves discontinuities well at the cost of oscillations near steep gradients. The neural formulations complement this picture rather than replace it. Strong-form PINNs reproduce the main flow features on the same benchmarks but remain several times

less accurate than the classical reference, with the primitive formulation consistently ahead of the conservative one. The finite-volume-informed PINN collapses to a spurious low-momentum state when trained on physics alone, and sparse data guidance removes the collapse monotonically with the amount of data supplied, although in the moderate-shock regime of these benchmarks its added discrete structure does not translate into an accuracy advantage over the simpler strong-form network. No neural method is competitive with the classical solver on a per-instance basis, which suggests that the practical payoff of physics-informed networks for dam-break flow lies in amortised, differentiable surrogates rather than in single-instance accuracy. The natural continuation is operator learning for cross-scenario generalisation [28, 29], together with wet/dry fronts, friction, and real topography.

References

- [1] Eleuterio F. Toro. *Shock-Capturing Methods for Free-Surface Shallow Flows*. John Wiley & Sons, Chichester, 2001.
- [2] Alexander Kurganov. Finite-volume schemes for shallow-water equations. *Acta Numerica*, 27:289–351, 2018.
- [3] Alfredo Bermúdez and María Elena Vázquez. Upwind methods for hyperbolic conservation laws with source terms. *Computers & Fluids*, 23(8):1049–1071, 1994.
- [4] Emmanuel Audusse, François Bouchut, Marie-Odile Bristeau, Rupert Klein, and Benoît Perthame. A fast and stable well-balanced scheme with hydrostatic reconstruction for shallow water flows. *SIAM Journal on Scientific Computing*, 25(6):2050–2065, 2004.
- [5] Peter D. Lax and Burton Wendroff. Systems of conservation laws. *Communications on Pure and Applied Mathematics*, 13(2):217–237, 1960.
- [6] Sergei K. Godunov. A difference method for numerical calculation of discontinuous solutions of the equations of hydrodynamics. *Matematicheskii Sbornik*, 47(89)(3):271–306, 1959. In Russian.

- [7] Philip L. Roe. Approximate Riemann solvers, parameter vectors, and difference schemes. *Journal of Computational Physics*, 43(2):357–372, 1981.
- [8] Amiram Harten, Peter D. Lax, and Bram van Leer. On upstream differencing and Godunov-type schemes for hyperbolic conservation laws. *SIAM Review*, 25(1):35–61, 1983.
- [9] Eleuterio F. Toro, M. Spruce, and W. Speares. Restoration of the contact surface in the HLL-Riemann solver. *Shock Waves*, 4(1):25–34, 1994.
- [10] Bram van Leer. Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov’s method. *Journal of Computational Physics*, 32(1):101–136, 1979.
- [11] Sigal Gottlieb, Chi-Wang Shu, and Eitan Tadmor. Strong stability-preserving high-order time discretization methods. *SIAM Review*, 43(1):89–112, 2001.
- [12] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [13] Susanna Dazzi. Physics-informed neural networks for the augmented system of shallow water equations with topography. *Water Resources Research*, 60:e2023WR036589, 2024.
- [14] Xin Qi, Gustavo A. M. de Almeida, and Sergio Maldonado. Physics-informed neural networks for solving flow problems modeled by the 2D shallow water equations without labeled data. *Journal of Hydrology*, 636:131263, 2024.
- [15] Yongfu Tian, Shan Ding, Lida Huang, Guofeng Su, and Jianguo Chen. Physics-informed neural networks for solving the two-dimensional shallow water equations with terrain topography and rainfall source terms. *Water Resources Research*, 61:e2025WR040052, 2025. Preprint: arXiv:2501.11372.
- [16] Kinza Mumtaz, Muhammad Waasif Nadeem, Adnan Khan, and Zahra Lakdawala. Investigating the use of physics informed neural networks for dam-break scenarios. *PLOS ONE*, 20(9):e0332694, 2025.

- [17] Xin Qi, Dawei Zhang, Fan Wang, Wuxia Bi, and Mingda Lu. NDAWL-PINN: a new non-dimensionalization and multi-task learning approach for efficient training of physics-informed neural networks to solve the shallow water equations. *Engineering Applications of Computational Fluid Mechanics*, 19(1):2535015, 2025.
- [18] Zhiping Mao, Ameya D. Jagtap, and George Em Karniadakis. Physics-informed neural networks for high-speed flows. *Computer Methods in Applied Mechanics and Engineering*, 360:112789, 2020.
- [19] Ismail Oubarka, Imad Kissami, Mohamed Boubekour, Fayssal Benkhaldoun, Aziz Madrane, and Zakaria Saadi. Weak and entropy physics-informed neural networks for conservation laws. arXiv preprint arXiv:2603.24819, 2026.
- [20] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred A. Hamprecht, Yoshua Bengio, and Aaron Courville. On the spectral bias of neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 5301–5310. PMLR, 2019.
- [21] Xiaofeng Liu. Finite volume-informed neural network framework for 2D shallow water equations: Rugged loss landscapes and the importance of data guidance. arXiv preprint arXiv:2605.11001, 2026.
- [22] Chang Wei, Yuchen Fan, Jian Cheng Wong, Chin Chun Ooi, Heyang Wang, and Pao-Hsiung Chiu. FFV-PINN: A fast physics-informed neural network with simplified finite volume discretization and residual correction. *Computer Methods in Applied Mechanics and Engineering*, 444:118139, 2025. Preprint: arXiv:2603.24114.
- [23] Dimitrios G. Patsatzis, Mario di Bernardo, Lucia Russo, and Constantinos Siettos. GoRINNs: Godunov-Riemann informed neural networks for learning hyperbolic conservation laws. *Journal of Computational Physics*, 534:114002, 2025.
- [24] Nathan Lichtlé, Alexi Canesse, Zhe Fu, Hossein Nick Zinat Matin, Maria Laura Delle Monache, and Alexandre M. Bayen. (U)NFV: Supervised and unsupervised neural

finite volume methods for solving hyperbolic PDEs. arXiv preprint arXiv:2505.23702, 2025.
Also on OpenReview, id AhtDnPyfOE.

- [25] Jingxiao Wu, Qiuhua Liang, Huili Chen, and Haoran Duan. RimNet: A hybrid neural network–finite volume framework for Riemann flux approximation in shallow water equations. SSRN preprint, abstract no. 5433524, 2025. Preprint posted September 2025.
- [26] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.
- [27] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.08895.
- [28] Zongyi Li, Hongkai Zheng, Nikola Kovachki, David Jin, Haoxuan Chen, Burigede Liu, Kamyar Azizzadenesheli, and Anima Anandkumar. Physics-informed neural operator for learning partial differential equations. *ACM/IMS Journal of Data Science*, 1(3):1–27, 2024.
- [29] Hao Wang, Chi-Wang Shu, and Qi Tang. LGNO: A local-global neural operator for hyperbolic conservation laws, 2026. arXiv:2606.18221.